

EC466/966: Estimation and Inference in Econometrics

Exercise # 2

1. Please answer the questions on EC466hw2.py. Please copy and paste **beauty.xls** file into the appropriate folder computer.
2. Suppose that you have the following regression model

$$\mathbf{y} = \mathbf{X}\beta + \epsilon,$$

where $\mathbf{y}$ is the $n \times 1$ vector of the dependent variable, $\mathbf{X}$ is the $n \times K$ vector of regressors, β is the $K \times 1$ vector of the model parameters, and ϵ is the $n \times 1$ vector of the error term. Assume that Classical Linear Regression Model assumptions hold. We estimate the model by the OLS estimator, and obtain $\mathbf{b}$ that is the estimate for β .

- (a) Show that $\hat{\mathbf{y}} = \mathbf{P}\mathbf{y}$ where $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.
- (b) Show that $\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}\epsilon$ where $\mathbf{M} = \mathbf{I}_n - \mathbf{P}$ and $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$.
- (c) Show that $SSR(\mathbf{b}) = \epsilon'\mathbf{M}\epsilon$.
- (d) Show that $s^2 = \mathbf{e}'\mathbf{e}/(n - K)$ is an unbiased estimator for $\sigma^2 = E(\epsilon_i^2|\mathbf{X})$.